

4.3 Option 1: Tectonic Activity and Hazards

Tectonic activity

Tectonic activity generates a wide range of natural hazards. The fundamental cause of these is plate tectonics, and thus the hazards have a distinct geography, linked to different tectonic settings. Tectonics is a key landscape-forming process which produces distinctive landforms in active regions, ranging from minor surface features such as faults or scarps to vast rift valleys and shield volcanoes.

Tectonic hazards

Tectonic hazards generate significant risk to human populations and their possessions, related to their vulnerability and the magnitude and frequency of hazardous events. Risk varies due to many factors including level of economic development, preparedness and education. Hazard impacts may be short term or long term. People respond to hazard risk in a variety of ways by, for instance, attempting to modify the hazardous event, through vulnerability or loss. Response depends on knowledge, technology and the availability of financial resources.

Global synoptic context

Key linkages

Places

Where are the tectonically active regions of the world?

People

Who is affected by tectonic activity and its associated hazards?

Power

Who is responsible for managing the risk associated with tectonic hazards?

Suggested teaching and learning

Mapping plate boundaries and recent hazardous events.

Examining the socio-economic profiles of two countries at different levels of development, that have been affected by recent hazardous events.

Using a news website to draw up a mind map of organisations and groups involved in a well-known hazardous event, eg the 2005 Kashmir earthquake.

1 Tectonic hazards and causes

Enquiry question: *What are tectonic hazards and what causes them?*

What students need to learn	Suggested teaching and learning
■ Tectonic hazards and disasters and what makes tectonic activity hazardous. There is a range of tectonic hazards associated with both volcanoes (lava, pyroclastics, ash, lahars, etc) and earthquakes (ground shaking, displacement, liquefaction, tsunamis, etc).	■ Exploring the difference between a tectonic event, a hazard and a disaster and investigating why some tectonic activity is hazardous.
■ Event profile of hazards, including frequency, magnitude, duration and areal extent.	■ Developing an awareness that hazards vary in their profile (frequency, magnitude, duration, areal extent, etc).

■ The causes of tectonic hazards, including the pattern of tectonic plates, their movements and possible causes of these movements.	■ Investigating the global pattern of plates and their movements and researching current theories explaining these movements, largely relating to convection currents in the asthenosphere.
■ Tectonic activity associated with different types of plate margins (convergent, divergent and transform) and the impact of this on the spatial distribution of tectonic hazards.	■ Developing an awareness that specific tectonic activity occurs at different plate margins and that this affects the spatial pattern of tectonic hazards.

2 Tectonic hazard physical impacts

Enquiry question: *What impact does tectonic activity have on landscapes and why does this impact vary?*

What students need to learn	Suggested teaching and learning
■ The varying impact of extrusive igneous activity, including the formation of volcanic cones, fissures and lava plateaux.	■ Developing an awareness as to why extrusive igneous activity varies from place to place and investigating why this produces a range of landscape impacts.
■ The formation and morphology of different types of volcano and the characteristics of different types of eruption. How these variations link to the processes at different plate margins and at hot spots.	■ Investigating differences in the morphology and eruption characteristics of different types of volcano.
■ The varying impact of intrusive igneous activity, both major and minor.	■ Examining the range of major and minor features of intrusive activity and how even intrusive activity can have impacts on the surface landscape features.
■ The effects that earthquakes can have on landscapes, such as fault lines, rift valleys and ground displacement, and the stresses and forces the ground is subjected to during such activity.	■ Researching the varying impact of earthquakes, including analysis of the stresses giving rise to faults.

3 Tectonic hazard human impacts

Enquiry question: *What impacts do tectonic hazards have on people and how do these impacts vary?*

What students need to learn	Suggested teaching and learning
■ The reasons why people live in tectonically active areas and how this might relate to the level of economic development.	■ Investigating the varying reasons why people live in tectonically active areas (such as ignorance, choice, inertia).
■ The range of hazards associated with different types of tectonic activity.	■ Weighing up a range of impacts that tectonic hazards have on people, including physical, economic and social impacts.
■ The specific impacts of a range of tectonic hazards at a range of scales and at locations in countries at different stages of development.	■ Researching the range of impacts of hazards in contrasting locations.
■ Trends in frequency and impact over time.	■ Developing an awareness of the varying patterns of impact over time, both long and short term (eg Park Model).

4 Response to tectonic hazards

Enquiry question: *How do people cope with tectonic hazards and what are the issues for the future?*

What students need to learn	Suggested teaching and learning
■ The varying approaches of individuals and governments to coping with tectonic hazards in countries at different stages of development.	■ Exploring the range of approaches that exist in attempting to cope with tectonic hazards, including do nothing, adjust and leave.
■ Specific strategies involved in adjustment: modifying loss burden, modifying the event and modifying human vulnerability; and the range of approaches and strategies used in locations at different stages of development.	■ Researching the different ways in which people attempt to cope, before, during and after events.
■ The effectiveness of different approaches and methods of coping and the way in which approaches have changed over time, and possible future coping strategies.	■ Investigating how successful coping strategies tend to be, undertaking cost-benefit analysis. ■ Developing an awareness of how strategies have changed and developed over time and how they might continue to change and develop in the future.

Suggested fieldwork opportunities for tectonic activity and hazards

Local fieldwork ■ Field visits to examine small-scale igneous structure and their impact on landscapes. ■ Faulting and rifting, and its impact on the landscape could be illustrated by field visits.	Residential and long-haul fieldwork ■ Many opportunities to study the nature of, and response to, hazardous events exist in Iceland, and the volcanic fields of Italy. ■ The impact of igneous activity on landforms can be seen in Arrow, Skye, the Lake District, North Wales and Northumberland etc.
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